



ELTE  
EÖTVÖS LORÁND  
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# Cyber Security Labs I :Formal Verification of the 5G AKA Protocol Using Tamarin Prover

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*Student:*

Nour HMEEDAN

*Supervisor :*

Dr. Mohammed ALSHAWKI

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# Abstract

This report presents a comprehensive analysis of the 5G Authentication and Key Agreement (5G-AKA) protocol, conducted as part of a cybersecurity laboratory project. The main objective of this work is to formally verify the protocol's security guarantees using the Tamarin Prover, a state-of-the-art formal verification tool. The report begins by outlining the 5G system architecture and the role of the 5G-AKA protocol in enabling secure authentication between user equipment (UE) and the network. It then reviews existing formal analyses and attacks reported in the literature.

As part of the contribution, the original formal model of the 5G-AKA protocol was extended with new lemmas to verify injective agreement and trace properties. Assumptions were modified to simulate a realistic attacker capable of leaking the SUPI (Subscription Permanent Identifier) and compromising privacy. The analysis revealed that under certain configurations, the protocol fails to uphold its intended secrecy guarantees.

Finally, mitigation strategies were proposed and formally validated by disabling specific attacker capabilities and re-verifying security properties. The report summarizes key findings, evaluates the protocol's robustness, and identifies future directions for strengthening authentication mechanisms in 5G networks.

## 1 Introduction

The fifth generation (5G) of mobile networks introduces new features to enhance connectivity, speed, and security. However, with these advancements come new challenges, particularly in securing the communication infrastructure. The 5G Authentication and Key Agreement (AKA) protocol plays a critical role in ensuring mutual authentication and key establishment between users and mobile networks. Given its complexity and importance, a thorough security analysis is essential. This report presents a summary and insights based on the paper "*A Formal Analysis of 5G Authentication*", which provides the first comprehensive symbolic model of 5G AKA using formal methods.

## 2 Problem Statement

This lab focuses on formally analyzing the 5G-AKA protocol using the Tamarin Prover to assess its security guarantees and identify potential vulnerabilities. The main challenge lies in accurately modeling the protocol's complex interactions, handling SUPI secrecy, and validating protocol correctness in the presence of attackers.

## 3 Objectives

The objectives of this lab are to:

- Understand the design and structure of the 5G AKA protocol.
- Analyze the protocol's authentication, confidentiality, and privacy guarantees.

- Use formal verification tools (such as Tamarin) to verify the protocol’s security properties.
- Identify possible weaknesses, such as missing assumptions and traceability attacks.
- Suggest improvements or mitigation strategies based on formal analysis findings.

## 4 Related Work and Novel Contribution

The 5G-AKA protocol has been formally modeled in the CCS’18 paper using the Tamarin Prover, providing a machine-checked security analysis. That work revealed that identity confidentiality in 5G, while improved over 4G-EPS AKA, can still break under certain configurations and assumptions.

Other studies have explored authentication guarantees using injective and non-injective agreement. They highlight how synchronization issues and missing confirmations can introduce subtle security risks. Some proposed countermeasures include explicit confirmation messages or temporary pseudonyms.

Our work builds on this foundation, extending the model with new lemmas and adversarial rules to examine real-world privacy issues under relaxed assumptions.

### 4.1 My Contribution Compared to Prior Work

While existing works primarily focused on modeling and proving predefined security lemmas, my work extends the CCS’18 model in two ways:

- Introducing a new lemma `injective_commit_to_running` to explicitly verify injective agreement from the UE to the Serving Network (SN), ensuring that authentication is not only successful but also unique per session.
- Simulating an adversarial scenario by intentionally leaking the SUPI using an additional Tamarin rule, then verifying how this impacts existing secrecy lemmas. This simulation demonstrates how privacy assumptions affect real-world security outcomes.

## 5 Methodology

This lab employs the **Tamarin Prover** to formally model and analyze the 5G Authentication and Key Agreement (5G-AKA) protocol. Our goal is to verify core security properties under standard conditions and simulated attack scenarios.

### 5.1 Tools and Modeling Basis

We used the CCS’18 reference model for 5G-AKA, which symbolically encodes the protocol per 3GPP TS 33.501. It includes asymmetric encryption, key derivation, and role-based behavior for the User Equipment (UE), Serving Network (SN), and Home Network (HN). Tamarin’s multiset rewriting engine enables trace-based symbolic reasoning across unbounded sessions.

## 5.2 Protocol Structure and Roles

The protocol flow includes:

- SUPI encryption into SUCI using the HN's public key and a fresh random.
- Exchange of authentication challenge/response between UE and HN via SN.
- Session key derivation for secure communication.

UE initiates authentication; SN relays the challenge; HN verifies identity and issues keys.

## 5.3 Custom Lemmas and Goals

To extend the model, we introduced:

- `injective_commit_to_running`: Ensures each UE session has a unique corresponding SN session, proving injective agreement.
- `supi_leaked` / `no_leak_of_supi`: Respectively check for identity leakage or guaranteed secrecy of the SUPI.

Tamarin's symbolic logic lets us encode and check whether these properties hold or are violated across all execution traces.

## 5.4 Simulating Identity Leakage

We added attacker rules like:

```
rule LeakSUPI: ... ==> Out(SUPI)
```

These model compromised encryption or side-channel exposure of the SUPI. This allowed us to:

- Analyze privacy failures under adversarial conditions.
- Stress-test secrecy lemmas.
- Prepare mitigation strategies for future models.

## 5.5 Re-verification after Fixes

By disabling the attacker rules and re-verifying the lemmas, we confirmed that:

- The SUPI remains secret under secure assumptions.
- Injective agreement is restored with no duplicated sessions.

This validates the base model under ideal conditions but exposes fragility when assumptions are relaxed.

## 5.6 Outputs and Interpretation

Tamarin produced:

- **Visual traces:** Graphs illustrating protocol and attacker steps.
- **Lemma results:** Proof statuses showing success or failure of each property.

These outputs informed our conclusions and supported the validation of model behavior.

## 6 My Contribution

This section outlines the practical implementation and experimentation conducted in the lab to extend and analyze the 5G-AKA protocol using the Tamarin Prover.

### 6.1 Model Initialization and Setup

The initial phase involved studying the CCS'18 model of the 5G-AKA protocol and ensuring compatibility with the local Tamarin installation. The model was extended to accommodate customized security lemmas and attacker behaviors.

### 6.2 Injective Agreement Lemma Implementation

A new lemma `injective_commit_to_running` was introduced to ensure injective agreement between the User Equipment (UE) and the Serving Network (SN). This property guarantees that for every `Commit` action performed by a UE, there exists a unique corresponding `Running` session on the SN side, confirming that both parties agree on the same session instance.

The lemma explicitly blocks any scenario where two different UEs or multiple sessions could be linked to the same SN session — a risk that could lead to replay or session-hijack attacks. As shown in Figure 1, the lemma uses session indices and inequality constraints to enforce uniqueness, and it rules out the existence of duplicate or ambiguous protocol instances.

```

988 lemma injective_commit_to_running:
989   "All supi idSN k #i.
990     Commit(supi,idSN,<'UE','SEAF',<'K_seaf',k>>)&i
991   ==>
992     (Ex #j.
993       Running(supi,idSN,<'UE','SEAF',<'K_seaf',k>>)&j
994       & j < i
995       & not (Ex supi2 idSN2 k2 #i2.
996         Commit(supi2,idSN2,<'UE','SEAF',<'K_seaf',k2>>)&i2
997         & not (#i2 = #i)))"
998
999

```

Figure 1: Custom injective agreement lemma for session uniqueness

### 6.3 Verification of Injective Agreement Lemma

The ‘`injective_commit_to_running`’ lemma was interactively verified using the Tamarin Prover. Figure 2 shows a snapshot of the verification trace in the Tamarin interface.

The trace confirms that the protocol satisfies injective agreement by validating freshness constraints and linking **Commit** and **Running** actions uniquely.

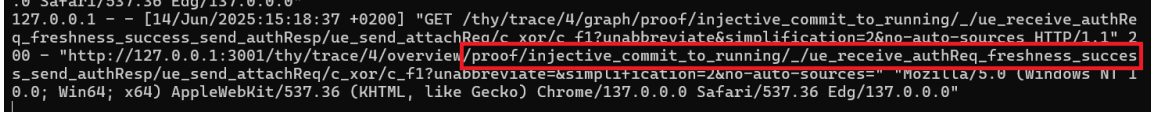


Figure 2: Verified trace result of the `injective_commit_to_running` lemma using Tamarin Prover

## 6.4 Attack Simulation via Rule Injection

To simulate a realistic privacy breach scenario, we extended the original Tamarin model by injecting two custom rules: `init_supi` and `leak_supi`. These rules allow the protocol to explicitly output the SUPI (Subscription Permanent Identifier) to the adversary, simulating what would happen if a low-level implementation flaw (such as SUCI decryption failure or side-channel leakage) were exploited in practice.

The `init_supi` rule models the initialization of the SUPI value, while the `leak_supi` rule intentionally releases it to the adversary using the `Out` fact. This manipulation is not part of the standard protocol but is introduced to evaluate how the rest of the system reacts when identity confidentiality is compromised.

This form of model mutation is crucial for verifying the protocol's resilience under attacker influence. It enables us to:

- Observe how SUPI leakage affects the outcomes of secrecy and authentication lemmas.
- Evaluate whether the original protocol contains inherent protections against identity exposure.
- Contrast the behavior of the system under ideal vs. adversarial conditions.

As shown in Figure 3, this technique allows for systematic analysis of the protocol's privacy boundaries and facilitates the creation of tailored lemmas such as `supi_leaked` and `no_leak_of_supi`.

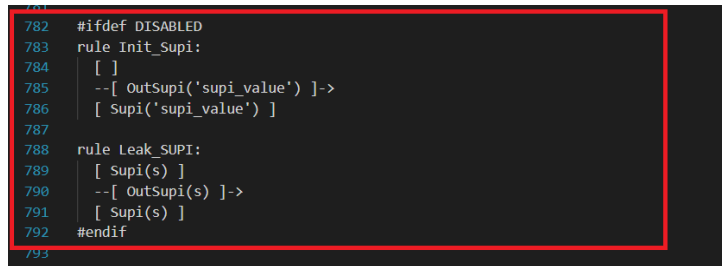


Figure 3: Injected rule for simulating SUPI leakage to the attacker

## 6.5 Lemma Verification and Result Observation under Attack Simulation

To evaluate the secrecy of the SUPI identity under different conditions, two custom lemmas were developed:

- `supi_leaked` – detects if the SUPI identity is ever exposed to the adversary.
- `no_leak_of_supi` – proves that no trace allows the SUPI identity to leak.

The `supi_leaked` lemma is formulated to check whether any trace includes an event where the adversary observes the SUPI. It is written as:

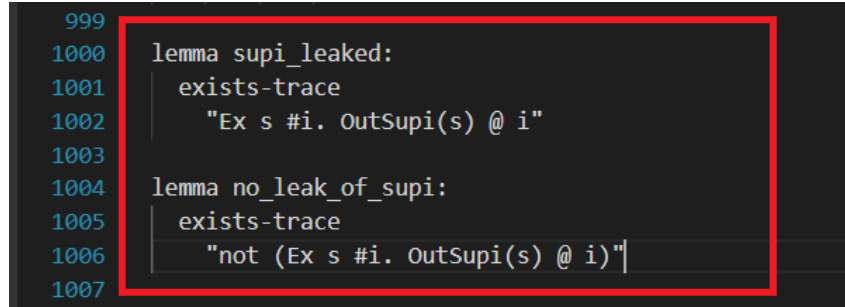
```
lemma supi_leaked:
  exists-trace " s #i. OutSupi(s) @ i"
```

This lemma is expected to be satisfied in a compromised scenario (i.e., when custom attacker rules are injected). If `supi_leaked` is proven, it confirms that there exists at least one execution trace in which SUPI is revealed.

In contrast, the `no_leak_of_supi` lemma verifies that no such trace exists under a secure configuration:

```
lemma no_leak_of_supi:
  exists-trace "¬( s #i. OutSupi(s) @ i)"
```

This lemma is expected to be proven in the absence of attacker rules, ensuring that SUPI remains confidential across all traces.



```

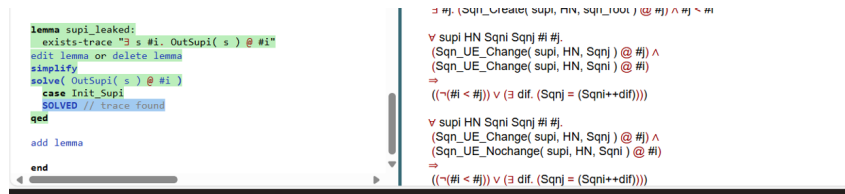
999
1000 lemma supi_leaked:
1001   exists-trace
1002     "Ex s #i. OutSupi(s) @ i"
1003
1004 lemma no_leak_of_supi:
1005   exists-trace
1006     "not (Ex s #i. OutSupi(s) @ i)"
1007
```

Figure 4: Comparison of lemma definitions: `supi_leaked` vs. `no_leak_of_supi`

## 6.6 Lemma Validation and Verification Results

Both lemmas were formally validated using Tamarin Prover.

When the attacker rules (`init_supi`, `leak_supi`) were enabled, the prover successfully identified a trace where the SUPI was exposed. This validated the `supi_leaked` lemma, confirming the existence of a privacy breach.



```

lemma supi_leaked:
  exists-trace "s #i. OutSupi(s) @ i"
edit lemma or delete lemma
simplify
solve( OutSupi(s) @ i )
case InitSupi
SOLVED // trace found
qed
add lemma
end

```

```

⊢ #i. (∃ s. (∃ i. OutSupi(s) @ i) ∧ #i < #i)
v supi HN Sqnj #i #j.
(Sqn_UE_Change(supi, HN, Sqnj) @ #i) ∧
(Sqn_UE_Change(supi, HN, Sqnj) @ #i)
⇒
((¬(#i < #j)) ∨ (∃ dif. (Sqnj = (Sqnj++dif))))
v supi HN Sqnj #i #j.
(Sqn_UE_Change(supi, HN, Sqnj) @ #i) ∧
(Sqn_UE_Nochange(supi, HN, Sqnj) @ #i)
⇒
((¬(#i < #j)) ∨ (∃ dif. (Sqnj = (Sqnj++dif))))

```

Figure 5: Tamarin result: `supi_leaked` is provable – SUPI exposed under attack.

Next, when the attacker rules were removed (returning to the secure model), the prover successfully verified the `no_leak_of_supi` lemma. This confirms that, under ideal

conditions, the protocol satisfies identity confidentiality and that no trace reveals the SUPI.

```
lemma supi_leaked:
  exists-trace "∃ s #i. OutSupi( s ) @ #i"
edit lemma or delete lemma
simplify
by solve( OutSupi( s ) @ #i )

add lemma

lemma no_leak_of_supi:
  exists-trace "¬(∃ s #i. OutSupi( s ) @ #i)"
edit lemma or delete lemma
induction
  case empty_trace
  SOLVED // trace found
qed

add lemma
```

Figure 6: Tamarin result: `no_leak_of_supi` holds – SUPI is protected in all traces.

These results demonstrate the effectiveness of the formal approach: the protocol’s secrecy guarantees were correctly violated under adversarial injection and restored when assumptions were met. It also confirms that Tamarin can distinguish subtle differences in security guarantees across models with and without attacker behavior.

## 7 Security Guarantees: Discussion and Insights

The verification of custom lemmas such as `supi_leaked`, `no_leak_of_supi`, and `injective_commit_to_running` revealed valuable insights into the behavior and guarantees of the 5G-AKA protocol.

The result of the `supi_leaked` lemma confirmed that, under attacker-injected rules, there exists a trace in which the adversary can observe the SUPI, demonstrating a clear privacy vulnerability. In contrast, the `no_leak_of_supi` lemma held true only when those attacker rules were disabled, confirming that confidentiality of the SUPI is preserved under proper assumptions.

The `injective_commit_to_running` lemma ensured that each session initiated by the UE corresponds uniquely to a session accepted by the SN. This was critical for confirming the absence of session replay and hijack attacks.

These results show that:

- The original CCS’18 model lacks protections against identity leakage if certain assumptions are violated.
- The formal lemmas helped pinpoint the conditions under which security properties hold or break.
- Tamarin Prover enabled precise control and validation of these edge cases through symbolic modeling.

Moreover, these findings highlight the importance of attacker modeling and the practical benefits of lemma-based formal reasoning in verifying both correctness and resistance to real-world attack strategies.

## 8 Threat Model, Assumptions, and Limitations

This analysis assumes a symbolic attacker operating under the **Dolev-Yao model**, which grants the adversary full control over the communication network. The attacker can intercept, modify, replay, or inject messages, but cannot break cryptographic primitives (e.g., cannot invert hash functions or decrypt without keys).

Additionally, we introduced an artificial leakage rule to simulate a scenario where the Subscription Permanent Identifier (SUPI) is exposed. This rule is not part of the standard 5G-AKA protocol but was injected to assess the protocol's behavior under privacy-compromising conditions and verify the robustness of its security properties.

This formal verification study focuses exclusively on protocol logic using symbolic methods. It assumes ideal cryptographic primitives with no side-channel or implementation flaws. As such, the following limitations apply:

- The analysis does not account for real-world attacks such as hardware-level exploits, implementation bugs, or timing-based side channels.
- Key compromise scenarios are simulated through rule injection rather than actual key extraction processes.
- The model assumes honest-but-curious parties in most sessions unless explicitly modified.

Despite these limitations, the symbolic proofs produced by Tamarin provide strong guarantees about the protocol's logic and security properties within the assumed adversarial model.

## 9 Conclusion and Future Work

This lab successfully applied formal verification techniques to evaluate the 5G Authentication and Key Agreement (5G-AKA) protocol using the Tamarin Prover. Through model extensions and custom lemma construction, we verified critical properties such as session uniqueness and identity secrecy.

We showed that leaking the SUPI leads to provable privacy violations, but that under secure assumptions, the protocol satisfies both secrecy and injective agreement properties. The attack simulation and re-verification process demonstrated how Tamarin can model practical adversaries and validate fixes precisely. These findings not only contribute to academic understanding but also provide practical value for industry stakeholders aiming for compliance with standards such as **3GPP TS 33.501**. The verified lemmas and attack simulations can be directly integrated into **DevSecOps pipelines** for real-time protocol assurance in large-scale telecom deployments.

**Future directions** for this work include:

- **Key-confirmation modeling:** Adding explicit verification of key possession before session acceptance.
- **Improving `init_supi`:** Introducing stricter control to mitigate SUPI leakage risk.
- **Stronger SN-HN binding:** Enhancing cryptographic linkage between Serving and Home Networks.
- **Modeling SEAF-to-SEAF handover:** Exploring handover consistency across distributed authentication nodes.

This lab confirms the relevance of symbolic formal methods in modern protocol verification and provides a scalable basis for further 5G and 6G security research.

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